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A semantic framework for chemical process digitalisation using ontologies

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ABSTRACT

Digitalisation holds promise for the transformation of modern chemical processes into intelligent manufacturing systems that are responsive to dynamic market demands and sustainability goals. To achieve interconnection of physical installations with their digital twins, we developed a semantic framework which enables systematic knowledge management and effective utilisation of plant data, serving as a backbone of intelligent manufacturing. Our approach leverages ontologies and agents built on a knowledge graph infrastructure to host the deployment of both first-principles and artificial intelligence (AI) models of chemical processes. Additional ontologies were designed for connecting models with industrial plants. We present an end-to-end demonstration of the knowledge graph-based digital twin system and illustrate this technology with a use case of anomaly detection in an industrial pilot plant for nanomaterials synthesis. The implementation includes data acquisition, secure communication protocols, cloud-hosted data storage, dedicated AI workflows, and a user interface with version control for both ontologies and models. The results demonstrate the effectiveness of the semantic framework in managing chemical process knowledge, linking plants to first-principles and data-driven models, and enabling the execution of complex AI-driven workflows.

1. Introduction

Advances in computing and automation, broadly termed "digitalisation technologies", are triggering a fundamental transformation in the chemical industry and manufacturing [1]. As envisioned by the Industry 4.0 concept, future chemical processes will link physical and digital realms via digitalisation, delivering smart decision-making, enhanced productivity, and agility [2–5]. Recent rapid development and deployment of sensors, databases, and computational devices have enabled measurement, collection, storage, and use of extensive chemical process data, thereby laying the foundation for chemical process digitalisation and the application of artificial intelligence (AI) [6,7]. Digitalisation also serves as a crucial strategy to attain sustainability objectives in chemical processes [8]. The immense potential of chemical process digitalisation is widely acknowledged in both academic and industrial sectors [9].

Yet, chemical process digitalisation faces several challenges, such as the existence of data silos, inconsistent data formats, insufficient standardisation, and restricted interoperability across different systems and platforms [10–13]. The integration of multi-source components, ranging from real-time sensor readings to AI-driven process models, requires a structured approach to ensure seamless communication, compatibility, and reusability [14]. To fully exploit the potential of digitalisation technologies in chemical processes, a highly interoperable digital infrastructure is essential for efficient data management, knowledge utilisation, and model application [15].

Such a digital infrastructure should facilitate the integration of industrial devices and computational models, thereby enabling applications such as predictive maintenance, process optimisation, supply chain management, digital twins, and overall efficiency improvement [16].

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Achieving these capabilities necessitates data and models being structured, shared, and reused effectively across various platforms and stakeholders [17]. In this context, FAIR data principles (Findable, Accessible, Interoperable, and Reusable) are vital for the representation of various components of chemical processes [18]. By ensuring data and models adhere to FAIR principles, organisations can promote data integration, boost knowledge discovery, and enhance collaboration.

Within a developed digital infrastructure, ontologies play a crucial role in offering a standardised, machine-readable structure that supports interoperability and reusability [19]. Through ontologies, digital assets can be systematically classified, linked, and enriched with contextual meaning as nodes and edges in a knowledge graph, providing shared semantics across organisations and domains [20]. Ontologies allow encoding of software agents as executable components to retrieve resources by navigating on the knowledge graph [21]. Information about different aspects of a process can be represented as detached entities while preserving interconnections [22].

Significant efforts have been undertaken to develop ontologies that formalise concepts and knowledge of chemical process engineering. A pioneering work in this field is OntoCAPE, developed by Morbach et al., which serves as a large-scale ontology aimed at balancing usability and reusability in computer-aided process engineering [23]. To establish a semantic database for chemical species, Pascazio et al. designed OntoSpecies, incorporating species identifiers, chemical and physical properties, chemical classifications and applications, spectral information, and the provenance and attribution metadata [24]. OntoSpecies provides a comprehensive and reliable source of chemical data that can be easily accessed by users. Another notable contribution is OntoKin, an ontology designed to capture both data and semantics of chemical reaction mechanisms, facilitating the simulation and understanding of chemical process behaviours, albeit currently limited to gas phase chemistry [25].

Despite these advancements, existing ontologies typically focus on narrowly defined domains, such as process engineering principles, chemical species, or reaction kinetics. A unified semantic framework for systematic digitalisation of practical chemical processes is still lacking [26]. Central to these chemical process-related ontologies are models, either physical, data-driven, or a combination of both, capturing the underlying physico-chemical principles governing chemical process dynamics [27]. For physical modelling, model knowledge is expected to be represented as mathematical formulations and deposited in a model ontology, which can be translated into executable codes for practical applications [28]. Likewise, real-world chemical processes should be formalised in process ontologies, with detailed information delineated for all process modules to ensure their comprehensive digital representation for applying both physical and AI models. Model knowledge can therefore be reflected in detailed chemical process setups and instantiated as runnable models [29].

In this paper, we develop a semantic framework for chemical process digitalisation using ontologies and demonstrate it with anomaly detection in practical implementation. The ontology technology is utilised to enable semantic consistency, variable tag matching, and model interoperability across the life cycle of industrial data ranging from generation to processing, storage, and application in anomaly detection. Anomaly detection is of particular importance for the chemical industry, as modern chemical plants are usually composed of multiple interconnected components, where a partial abnormality may propagate through the whole system and lead to severe consequences, such as accidents, economic losses, and environmental damage [30]. Up to now, procedures dealing with non-normative plant behaviour remain heavily reliant on human operators, with approximately 70% of process accidents attributed to human errors [31]. This underscores the urgent need for intelligent systems for abnormal situation management that can effectively detect anomalies and prevent accidents.

Anomaly detection models can be broadly categorised into physical and data-driven models [32]. Data-driven models are prevailing due to their ease of development from plant data. Typical data-driven models include principal component analysis, kernel principal component analysis, partial least squares, and other methods [33–35]. Recent advances in deep learning have also prompted the investigation of neural networks in anomaly detection [36,37]. On the other hand, physical models can be favoured for their ability to express process knowledge in deterministic mathematical models, delivering robust applicability [38]. However, first-principles models are often difficult to develop and maintain. The complementary advantages of both types of models underscore the necessity of a framework that can leverage their synergy within a uniform digital infrastructure for anomaly detection.

In this work, respective ontologies were developed to represent model knowledge and process knowledge. Chemical process units, along with their interconnections and associated process variables, were delineated within the process ontology with linkages to the model ontology. Mass and gas-liquid phase transition balances were exemplarily captured in the model ontology. Corresponding agents were developed to automate subprocess identification, model construction, model calibration, and model application. Meanwhile, the process ontology plays the role of hosting and using AI models. Autoencoder (AE) was adopted as an example AI model for anomaly detection.

The remainder of the paper is organised as follows: Section 2 describes the overall idea of the current work with the developed underlying ontologies and applications. The implementation of the architecture is also described in this section. Section 3 first showcases the model maintenance and communication. Then, the section ends with the overall case studies of an in silico multi-unit chemical process and an actual industrial multi-mode dosing module. Finally, conclusions and prospects are summarised in Section 4.

2. Methods

2.1. An overview of the use of ontologies in digital chemical processes

Ontology creates explicit semantic mapping of physical reality, such as a laboratory experiment or an industrial plant, and their models, whether these are data-driven or first-principles or hybrid models; this is illustrated briefly in Fig. 1. The overall architecture consists of three layers: the ontology definition layer, the schematic representation layer, and the contextual agent layer.

The bottom ontology definition layer depicts related concepts, relations, attributes, and data types encapsulated in ontologies of different domains. In detail, concepts are represented as nodes in an ontology, with their internal connections defined as relations. Each concept is attached with a table indicating its attributes, as well as corresponding data types. In our framework, process ontology is developed to capture the chemical process superstructure composed of unit operations, streams, and process variables; model ontology encompasses common laws and parameters at the modelling knowledge level. Existing ontologies, such as DEXPI and Ontology of units of Measure (OM), are referred to for compatibility with common data exchange standards and unit definitions [39,40].

The middle semantic representation layer consists of interlinked instances of two ontologies – model ontology and process ontology – and the AI model, which together form the foundation of chemical process digitalisation. Taking a hydrodealkylation process as an example, instances of the process ontology are set to capture essential process elements, including unit operations, streams, and process variables, serving as a digital replica of the real-world hydrodealkylation setup. Fundamental physico-chemical laws and parameters are encoded as instances of the model ontology. To bridge these two, process variables of the process ontology are linked to parameters of the model ontology to indicate their physical entities in modelling. In addition, the process ontology can provide the context for applying AI models and offer a unified entry for aligned process data from sensors.

At the top layer are a series of contextual agents which drive

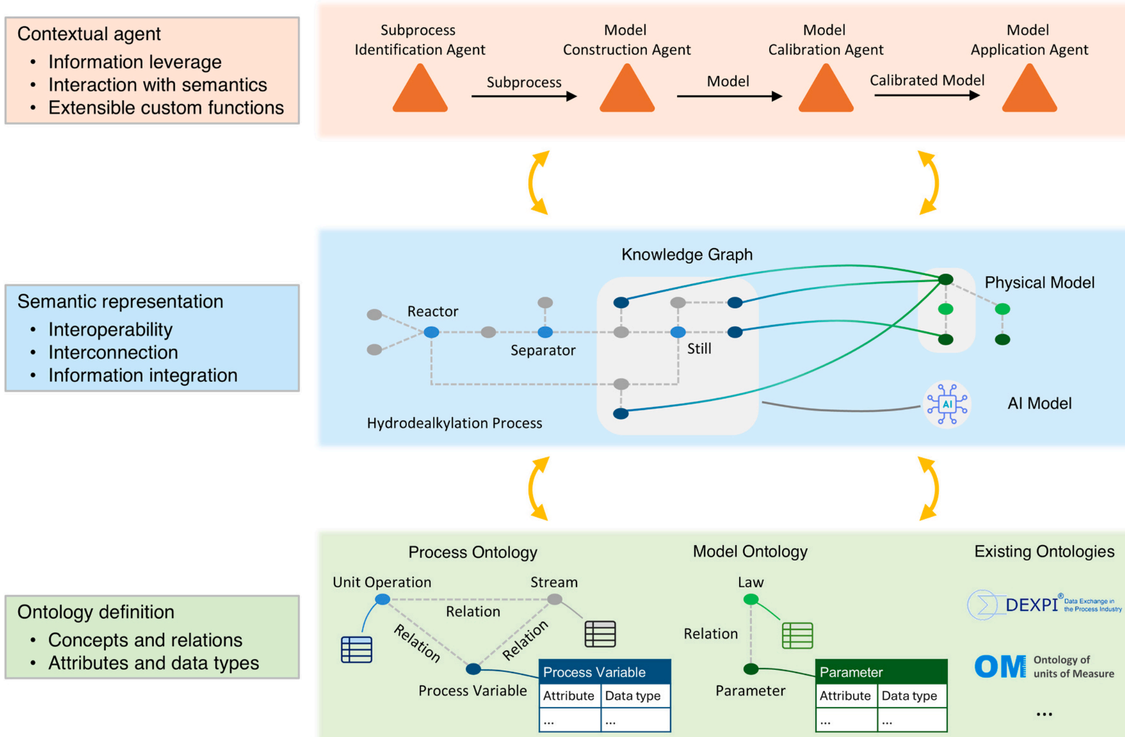

Fig. 1. Schematic illustration of the use of ontologies for digitising chemical processes.

chemical process modelling, including the subprocess identification agent, which recognises subprocesses and associates them with applicable laws (Fig. S1); the model construction agent, which generates physical models tailored to identified subprocesses (Fig. S2); the model calibration agent, which fits model parameters with process data; and the model application agent, which executes calibrated physical models and AI models for online process monitoring (Fig. S3). Together, these collaborative agents realise automated modelling, minimise human intervention, and enhance process monitoring adaptability.

By integrating these three layers, the proposed method establishes a scalable and adaptable way for chemical process digitalisation, offering a viable solution towards Industry 4.0 and smart manufacturing.

2.2. Metagraphs of model and process ontologies

Fig. 2 shows metagraphs of model and process ontologies, which

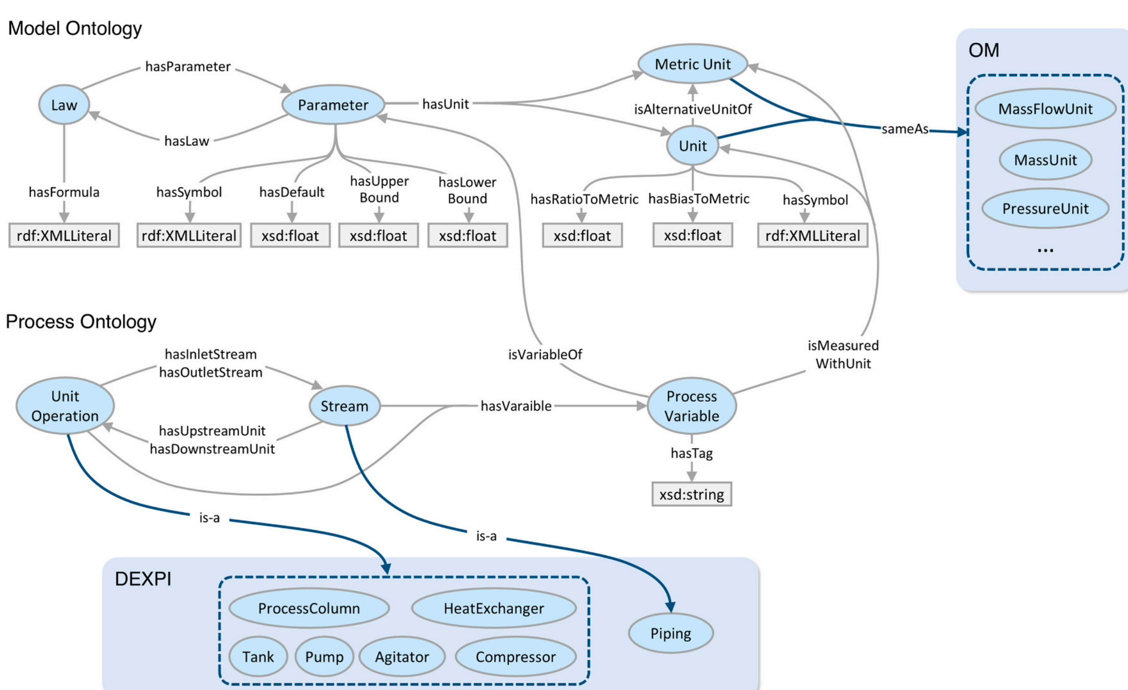

Fig. 2. Metagraphs of model and process ontologies used to digitalise chemical processes, illustrating defined classes, relationships, and associated data properties.

serve as the guidance for populating and extending ontologies. The model ontology is designed to represent general concepts and their relationships in physical modelling of chemical processes, encompassing *Law*, *Parameter*, and (*Metric*) *Unit*: *Law* defines the physics governing chemical processes, attached with Mathematical Markup Language (MathML) describing mathematical structures [41]; *Parameter* denotes the state and property of chemical process; (*Metric*) *Unit* is connected to *Parameter* to indicate its physical dimension and scale in measurement. Upper and lower bounds are introduced as properties of *Parameter* for model calibration. *Unit* is linked to *Metric Unit* for unification during calculation, with the ratio and bias noted in the model ontology.

The process ontology focuses on the representation of practical chemical processes: *Unit Operation* and *Stream* are defined to describe the superstructure with interconnections (*hasInletStream*, *hasOutletStream*, *hasUpstreamUnit*, and *hasDownstreamUnit*). *Pipe Manifold* is connected to *Stream* but omitted in the figure for clear elaboration. *Process Variable* is attached to *Unit Operation* and *Stream*, with edges linked to *Parameter* and (*Metric*) *Unit* to indicate its physical entity in modelling and measurement. The introduced tag string reveals the position of the measured variable in the chemical process diagram. *Unit Operation* and *Stream* are also mapped to the corresponding concepts defined in DEXPI to follow common data exchange standards in the process industry; *Unit* and *Metric Unit* are linked to their equivalents in OM for consistent semantic representation of units.

The integration of model and process ontologies facilitates the digital representation of chemical processes. By linking *Process Variable* to *Parameter*, this ontological method enables automated instantiation of physico-chemical laws in the physical setup to support process monitoring. The ontologies developed in this work can be found in Tables S1-S4.

2.3. Ontology-enabled physical modelling for process monitoring

Fig. 3 illustrates the ontology-enabled physical modelling for mass and gas-liquid phase transition balances, of which the formulas can be written as shown in Eqs. 1 and 2.

$$\int \sum Q_m I + \sum m[0] - \sum m[t] = 0 \quad (1)$$

$$\ln P + \frac{\Delta H}{R} \cdot \frac{1}{T} + C = 0 \quad (2)$$

where Q_m is the mass flow rate, I is the flow direction, m is accumulated mass, t is the time window size, P is pressure, T is temperature, ΔH is enthalpy change, R is the universal gas constant, and C is the intercept of the Clausius-Clapeyron equation [42]. The model construction agent retrieves these two formulas, together with candidate sub-formulas, from the model ontology and converts them into runnable codes for the identified subprocesses. These physical laws are stored in the form of MathML to represent the mathematical structure and relationships with parameters. In this work, physical models are ordinary differential equations (ODEs) implemented as Python functions for demonstration; meanwhile, it is notable that this workflow can be easily extended to other coding languages. A "MathML2Code" module has been developed to convert MathML to Python codes, as shown in Fig. S2.

Once the physical model is constructed, the model calibration agent can be activated to fit model parameters using process data, which are gathered from deployed databases according to process variable tags defined in the process ontology, as illustrated in Fig. S3. The SciPy library is adopted to solve the ODE and minimise the calculated mean squared error. After that, the physical model is aligned with the real-world process to ensure its usability in process monitoring. This workflow uses the differential evolution algorithm from SciPy to carry out parameter fitting [43]. The parameter settings for differential evolution are listed in Table S5.

2.4. System implementation

2.4.1. The overall system architecture

Fig. 4 illustrates the overall system architecture of the digital twin system, which is characterised by the utilisation of ontologies. Model ontology, process ontology, and other associated ontologies are managed in a knowledge graph database for data streaming, model construction, and anomaly detection for industrial applications. This ontology-driven system architecture ensures semantic consistency, interoperability, and reusability across process setup, time-series data, and developed models. With regard to the detailed implementation, two virtual machines, *Edge Virtual Machine* (*EdgeVM*) and *Agent Virtual Machine* (*AgentVM*), host the agents and software components. Cloud-based services, including *Gremlin GraphDB*, *MongoDB*, and *PostgreSQL*, are employed for data storage. *Gremlin GraphDB* stores ontology data. *PostgreSQL* manages intermediate computational results in relational tabular format. *MongoDB* stores time series data for efficient retrieval and analysis. Data transition rates of our architecture are summarised in Table 1. The hardware requirements for real plant deployment can be found in Table S6.

2.4.2. Data connectivity from plant to edge

The pilot plant case provides operational data essential for developing and evaluating the anomaly-detection system. Ontologies are constructed to formally represent selected devices, their associated parameters, and process data within a knowledge graph, enabling semantic integration and advanced analytics. To enable semantic integration within the digital twin system, a suite of ontologies was developed and extended to represent selected devices, associated parameters, and data from the dosing, reactor, and purification modules of the pilot plant (Figs. S4-S6). Ontology development was carried out using a layered approach, informed by plant process flow diagrams and expert input from process engineers. The ontologies reuse and extend concepts from existing domain ontologies: OntoCAPE for chemical process engineering [23], OntoDevice for representing physical devices [44], OntoBMS for building management systems [44], Ontology of Measure for units and quantities [40], SAREF for smart devices and functions [45], s4bldg for building-related concepts [46], and OntoSpecies for chemical species

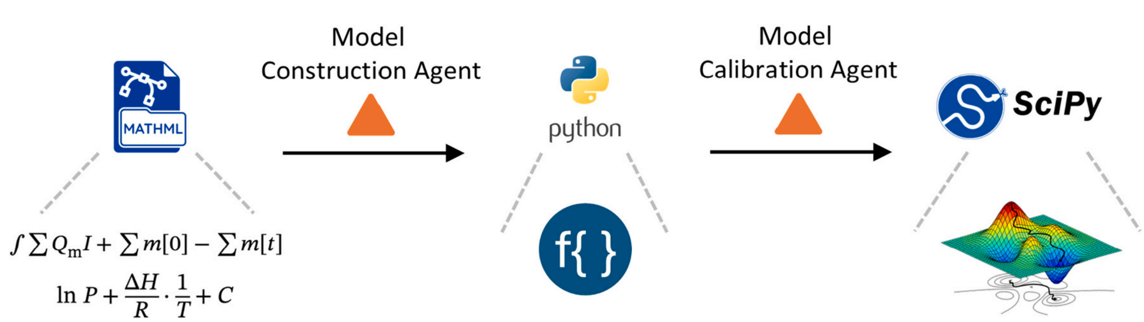

Fig. 3. An illustration of the ontology-enabled physical modelling for mass and gas-liquid phase transition balances.

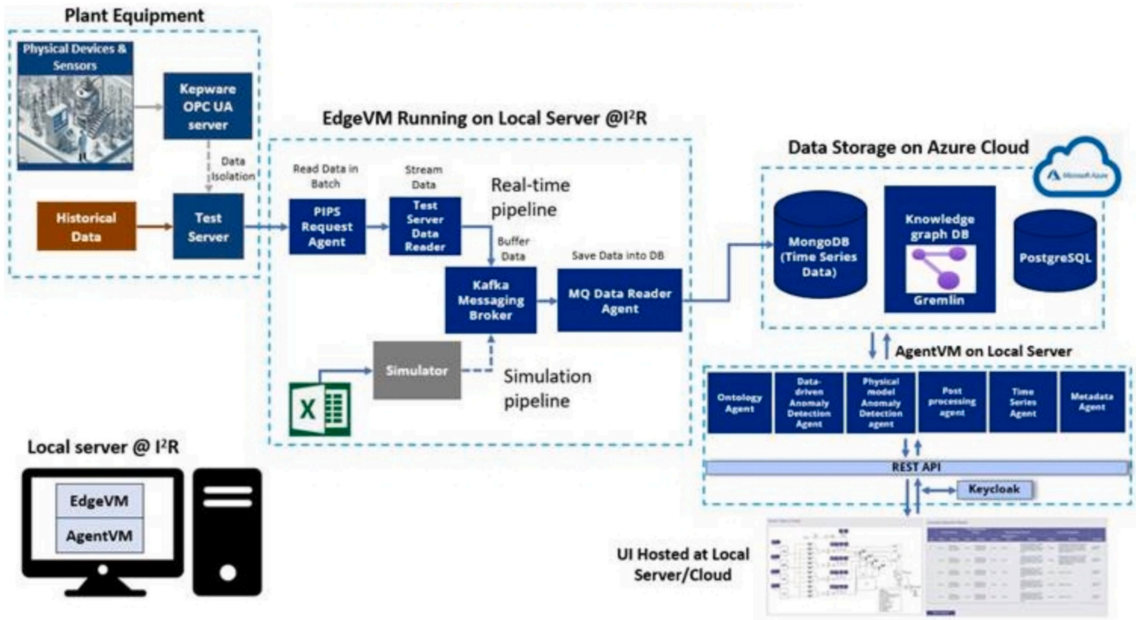

Fig. 4. The overall architecture for implementation of the ontology-based digital twin system.

Table 1
A summary of data transmission rates based on the developed architecture.

Data Flow	Data transmission rate	Time granularity (s)
'Plant Equipment' to 'EdgeVM'	3 Mbps	1[‡]
'EdgeVM' to 'Azure Cloud'	10–20 Mbps	1[‡]
'Azure Cloud' to 'Local Server/ Cloud'	1–3 Mbps	2–3
Components within 'Plant Equipment'	50 Mbps	50
Components within 'EdgeVM'	1 Gbps	2–3
Components within 'Azure Cloud'	50–100 Mbps	2–3

[‡] This data frequency can be adjusted and changed according to the requirement and the system capabilities.

[24]. The concepts from these ontologies are interconnected to enable alignment and data integration across multiple domains.

For example, a flow meter device in the dosing module is modelled using the *FlowMeter* class from OntoDevice. Its measured output, the volumetric flow rate, is linked to a quantity *VolumetricFlowRate* defined in OM, together with its associated values. The device's operational status may be connected to control logic using terms from OntoBMS, while metadata such as its physical location or function in the plant is described using s4bldg and SAREF concepts. Fig. 5 displays a part of the ontology developed to describe a flow meter and its related concepts from the dosing module. This layered integration ensures consistency and supports data integration across the digital twin system.

A data streaming architecture is developed to enable secure and efficient transfer of operational data from the pilot plant's Kepware Open Platform Communications Unified Architecture (OPC-UA) server to an Azure Internet of Things (IoT) Edge device. This architecture is designed to facilitate communication between the plant and external analytics platforms. The system integrates a combination of existing software components from the pilot plant and additional software agents, scripts, and libraries developed specifically for the data transfer pipeline.

As illustrated in Fig. 6, data from various devices and sensors in the pilot plant are first collected via a Programmable Logic Controller (PLC) and made accessible through the Kepware OPC-UA server. The *AMPLE Client* then automates the process of reading these data points by parsing a CSV file that defines all OPC-UA tags to be read. It processes the incoming timeseries and inserts it into a PostgreSQL database hosted on an Ubuntu server over a secure intranet connection. While most devices and sensors in the pilot plant communicate through the PLC, an exception is the mass flow controller, which uses proprietary software to read data. For this, a custom *OPC-UA Write* script has been developed to parse and forward the data to the Kepware server, after which the *AMPLE Client* continues the process of storing it in the database. Following local storage, the data is securely transmitted over the internet to the Azure IoT Edge device, demonstrating the feasibility of remote and secure access to pilot plant data.

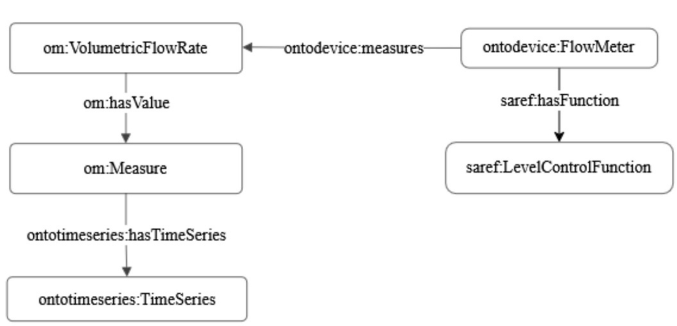

Fig. 5. A snippet of the ontology developed to describe a flow meter and the associated concepts.

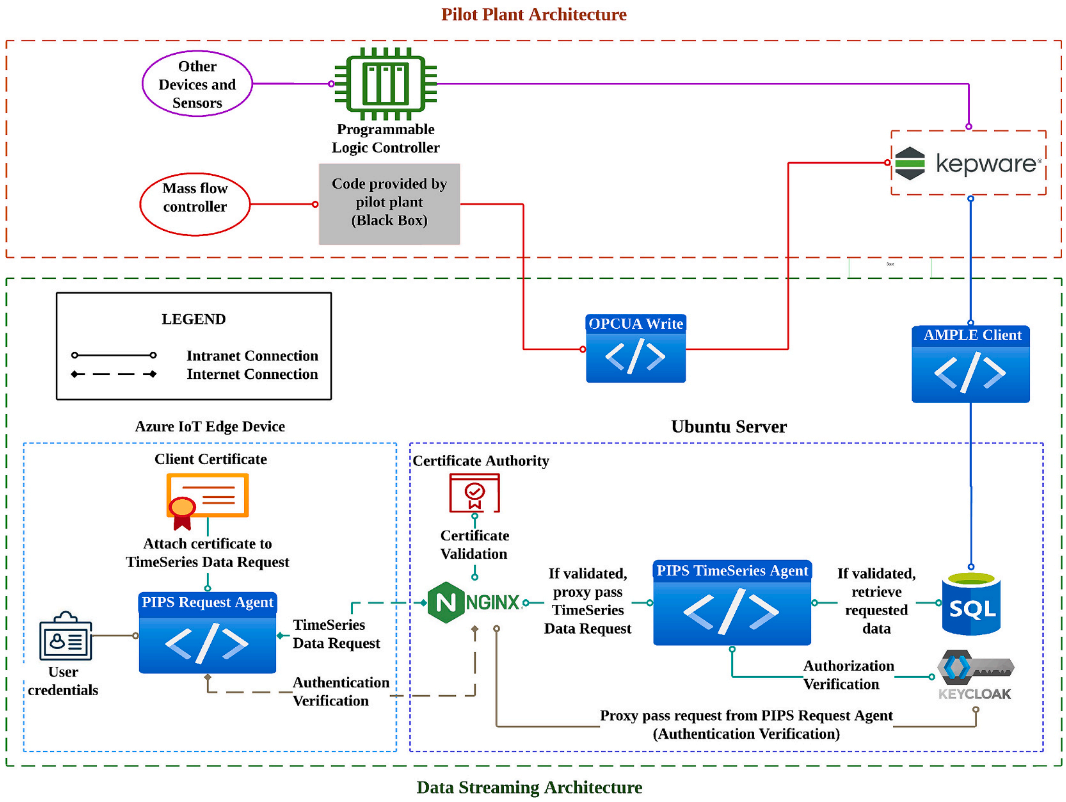

Fig. 6. An illustration of the data streaming architecture connecting a Kepware OPC-UA server to an Azure IoT Edge device. Two primary mechanisms have been implemented: Keycloak for authentication and authorisation, and Nginx as a reverse proxy for client certificate validation.

The data retrieval process begins with the *PIPS Request Agent*, which takes the user's authentication credentials as input and sends this to the *PIPS TimeSeries Agent* in a Hypertext Transfer Protocol Secure (HTTPS) request. If the credentials are valid, the *PIPS TimeSeries Agent* obtains the time-limited authorisation token from Keycloak, which is returned to the *PIPS Request Agent*. This token, along with a client certificate and details such as the number of recent readings to retrieve, is then included in the outgoing HTTPS request sent by the *PIPS Request Agent* for data retrieval.

The *PIPS TimeSeries Agent* serves as an Application Programming Interface (API) for retrieving timeseries from the PostgreSQL database. It accepts HTTPS requests specifying the data source, the number of recent readings to retrieve, and the authorisation token. Once the token and certificate are validated by Keycloak and Nginx, respectively, the *PIPS TimeSeries Agent* responds with a JavaScript Object Notation (JSON) object containing the requested timeseries. Unified Modelling Language (UML) diagrams of the entire process can be found in the Supplementary Information (Figs. S7-S9).

2.4.3. Edge processing, data storage and agents

The architecture as shown in Fig. 6 ensures seamless integration of data streams, AI-driven anomaly detection, and ontology-powered reasoning, ultimately enhancing system reliability, efficiency, and scalability for industrial applications. The following provides a brief overview of each component within the system architecture:

1) **Plant Equipment Layer**

   Plant Equipment hosts the *Kepware OPC-UA Server* and *Test Server*, which collect sensor data from the plant machinery and provide data access to the *PIPS Request Agent*.

2) **Edge VM**

   PIPS Request Agent is designed to receive data from the *Test Server* with *Keycloak* authentication and provides an *API* to read sensor data in batches.

   Test Server Data Reader Agent retrieves sensor data from the *PIPS Request Agent* and transmits it to the *Kafka Messaging Broker*, where the data is temporarily queued and buffered.

   MQ Data Reader Agent consumes the buffered data from the *Kafka Messaging Broker* and sends it to the *MongoDB* database for persistent storage and further analysis.

   The Simulator generates both healthy and unhealthy sensor data and sends it to the *Kafka Messaging Broker*, enabling the *Physical Model Agent* and the *Data-Driven Anomaly Detection Agent* to perform anomaly detection and error prediction.

3) **Agent VM**

   Ontology Agent allows users to upload ontology through a web-based interface, storing it in the *Gremlin* knowledge graph database. Additionally, it enables users to upload pre-trained models for ontology. The ontology and models are utilised by the *Physical Model Agent* and the *Data-Driven Anomaly Detection Agent* for anomaly detection.

   Physical Model Anomaly Detection Agent utilises embedded ontologies for physical models and equipment configurations, as well as sensor data for anomaly detection (Fig. S10(a)).

   Data-Driven Anomaly Detection Agent retrieves sensor data from the *MangoDB* database and the pre-trained model to perform real-time anomaly detection and root cause identification (Fig. S10 (b)).

   Post Processing Agent aggregates and fuses results from both *Physical Model Agent* and *Data-Driven Anomaly Detection Agent* to provide a comprehensive anomaly and root cause analysis outcome. See Fig. S11 for further details on anomaly detection.

   Time Series Agent reads time-series data from *MongoDB* and provides *API* endpoints for the web interface, enabling users to monitor real-time sensor readings and detected anomalies.

   Metadata Agent provides *API* endpoints for the web interface, enabling users to manage K100 experiment metadata.

Keycloak provides authentication and authorisation for *REST APIs*, ensuring secure access control and identity management.

User Interface allows operators to visualise real-time data monitoring dashboards and anomaly detection results, as well as to update and manage ontology, data-driven anomaly detection model, and experimental metadata.

For the data storage, three types of databases – *GraphDB*, *MongoDB*, and *PostgreSQL* – are utilised in the implementation to serve specific purposes based on the nature of the data. *GraphDB* is used to store and manage the ontologies in a graph structure. This is particularly beneficial for supporting ontology-driven applications such as *Physical Model Anomaly Detection Agent*. *MongoDB*, a *NoSQL* document-based database, is well-suited for storing large volumes of time-series sensor data due to its flexibility and ability to store unstructured or semi-structured data efficiently. *PostgreSQL*, a relational database, is used for storing structured metadata, such as ontology versions, data-driven anomaly detection agent models and their configurations. Using all three databases together allows the system to efficiently handle diverse data needs with improved performance, flexibility, and robustness.

An example user interface (UI) was set up to provide a seamless web-based experience for interacting with data and system agents. The developed UI integrates *APIs* from the timeseries agent, ontology agent, and metadata agent, allowing users to view real-time sensor data and detected anomalies through a dashboard, as well as upload and manage ontologies and anomaly detection models. This UI provides a user-friendly platform with secure login access for authorised users. It offers real-time display of sensor readings alongside anomaly detection results, which are presented both graphically and in tabular form. The table includes detailed information such as the timestamp of the faulty event, the final anomaly detection result, and the potential root cause of the fault. Users can download historical anomaly detection results for further analysis, upload and store metadata, and manage version control for both ontologies and anomaly detection models. This includes capabilities to upload, delete, and activate new versions, ensuring flexibility and traceability in system configuration and model management. More detailed examples of screenshots for various functions in the user interface are shown in Figs. S12-S20.

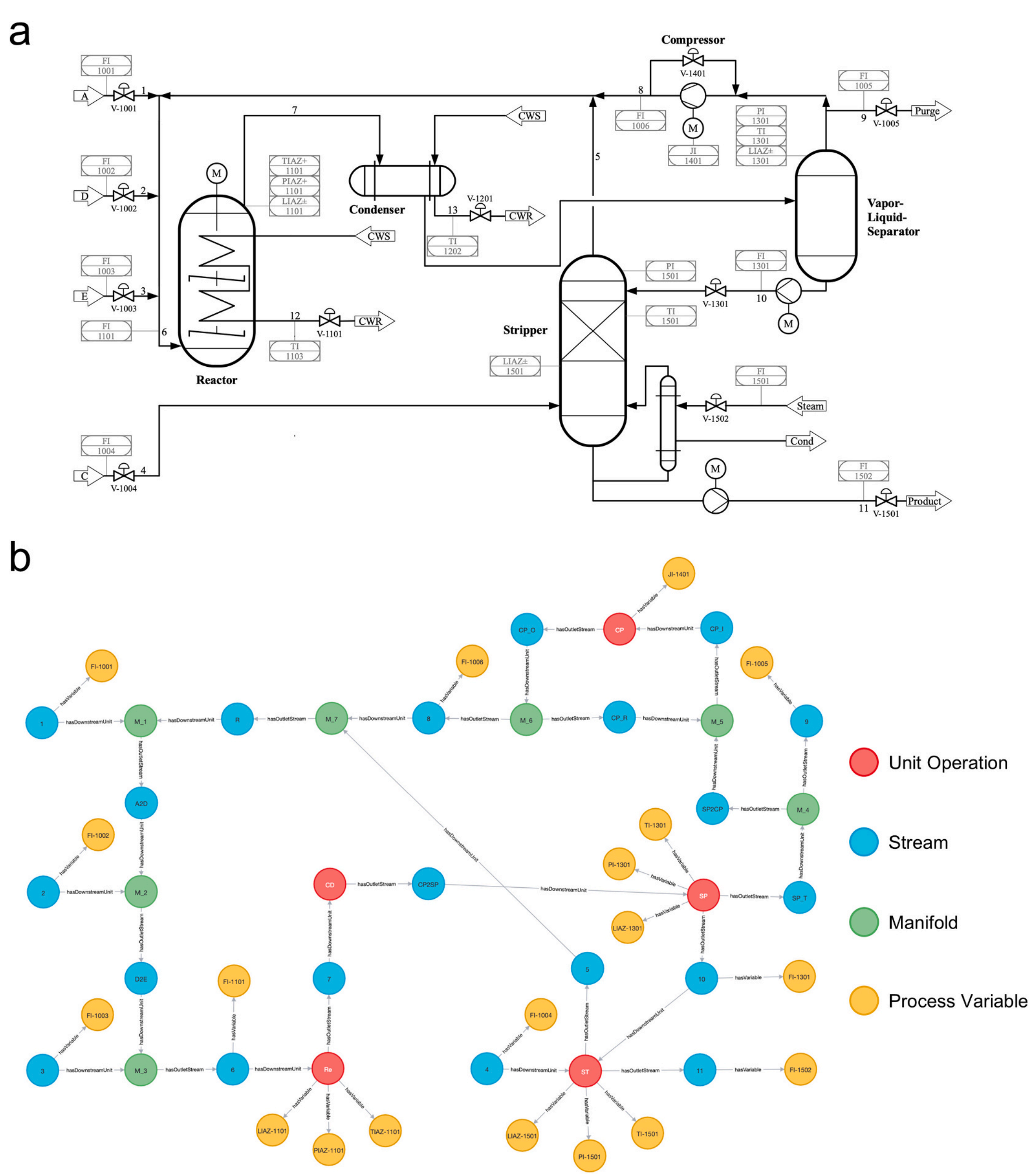

Fig. 7. Representation of the TE process as: (a) P&ID, and (b) a network representation.

3. Case studies

3.1. An in silico multi-unit chemical process case

3.1.1. Tennessee Eastman process

We first demonstrate our ontological method in silico using the multi-unit Tennessee Eastman (TE) process (Fig. 7(a)), a widely recognised benchmark for chemical process monitoring. Originally modelled by Downs and Vogel, the TE process model simulates an industrial chemical system characterised by gas-liquid phase transition and a recycle [47]. It produces two liquid products from four gaseous reactants through a series of exothermic reactions. The plant consists of five process units: a reactor, a condenser, a separator, a compressor, and a stripper. Fig. 7(b) visualises its process ontology using Neo4j, where unit operations are represented as red nodes, streams as blue nodes, pipe manifolds as green nodes, and process variables as yellow nodes [48]. For clarity, only *hasOutletStream* relationships between unit operations and streams are plotted. Given the extensive research on AI-based process monitoring for the TE process, the case study in this work is focused on physical model development using the developed ontologies.

The TE process involves 22 process variables, including temperature, pressure, flow rates, etc., which are recorded every three minutes (see Table S7 for details). To generate the dataset, the TE process is simulated for 48 h under normal and faulty operational conditions. A total of 15 faults with known root causes are separately introduced to the TE process after the simulation has been run under the normal operational condition for 8 h. Table 2 lists these faults alongside their types, descriptions, and disrupted balances. These faults can be categorised based on the process:

i) Controllable faults. Faults 3, 9, and 11 affect the reactor temperature and can be compensated by adjustments in the reactor cooling water flow rate by control loops. Faults 14 and 15 are caused by the cooling water valve sticking in the reactor and condenser, respectively, only leading to slight temperature fluctuations.
ii) Back-to-control faults. As step disturbances, faults 4, 5, and 7 deviate the process from the normal operational condition temporarily, of which the fault alarms should be eliminated once the process dynamics recover.
iii) Uncontrollable faults. Other faults exceed the capability of the control system and are considered uncontrollable [49].

Table 2
A description of the types and disrupted balances of TE process faults.

Fault	Type	Description	Disrupted Balance	
			Mass	Gas-liquid Phase Transition
1	Uncontrollable	A/C Feed Ratio (step)	✓	✓
2	Uncontrollable	B Composition (step)	✓	✓
3	Controllable	D Feed Temperature (step)	×	×
4	Back-to-control	Reactor Cooling Water Inlet Temperature (step)	×	×
5	Back-to-control	Condenser Cooling Water Inlet Temperature (step)	×	×
6	Uncontrollable	A Feed Loss (step)	✓	✓
7	Back-to-control	C Header Pressure Loss (step)	×	×
8	Uncontrollable	A, B, C Feed Composition (random variation)	✓	✓
9	Controllable	D Feed Temperature (random variation)	×	×
10	Uncontrollable	C Feed Temperature (random variation)	×	×
11	Controllable	Reactor Cooling Water Inlet Temperature (random variation)	×	×
12	Uncontrollable	Condenser Cooling Water Inlet Temperature (random variation)	×	✓
13	Uncontrollable	Reaction Kinetics (slow drift)	×	✓
14	Controllable	Reactor Cooling Water Valve (sticking)	×	×
15	Controllable	Condenser Cooling Water Valve (sticking)	×	×

3.1.2. Mass and gas-liquid phase transition balance modelling

Fig. 8 illustrates the automated mass balance modelling enabled by the developed ontologies method for the TE process. Hierarchical *Laws* are retrieved from the model ontology for modelling the mass balance, as shown in Fig. 8(a). The mass accumulation m is determined to be calculated as $\rho V_{\max} L$ by the subprocess identification agent, since the liquid volume is measured with the relative liquid level L. Fig. 8(b) shows the specific mass balance model for the TE process. The entire process is identified to be indivisible for mass balance calculations. The related flow indicators (FI-1001 to FI-1005, FI-1502) and level indicators (LIAZ-1101, LIAZ-1301, LIAZ-1501) are gathered as the model inputs. The derived model is then calibrated by optimising the density parameter ρ using process data under the normal operational condition. Subsequently, absolute errors of samples in the normal operational condition are calculated. To achieve robust anomaly detection, the time window is set to as large as 3 h for the absolute error calculation of mass and gas-liquid phase transition balances. Meanwhile, following the conventional practice, the alarm threshold is determined as the mean value plus four standard deviations.

The gas-liquid phase transition balance for the TE process is modelled in a similar pattern (Fig. 9(a)). In contrast to mass balance, where the entire process is indivisible for modelling, the subprocess identification agent reveals that the gas-liquid phase transition balance *Law* should be separately applied to the reactor, the separator, and the stripper. No sub-formulas are retrieved for gas-liquid phase transition balance, as all parameters are either process variables or parameters to be fitted. After that, model construction and calibration agents follow up to build gas-liquid phase transition balance models and determine alarm thresholds (Figs. 9(b), (c) and (d)).

3.1.3. Process monitoring results

Fig. 10 shows process monitoring results for controllable faults 3, 9, and 11. These three faults primarily disrupt the phase transition balance while imposing minimal impact on the mass balance. The resultant absolute errors confirm the effectiveness of the control system in handling these controllable faults. The monitored absolute error remains consistently below the alarm threshold for the mass balance and only occasionally exceeds the alarm threshold for the gas-liquid phase transition balance. Unlike conventional statistical models, the ontology-based process monitoring models are informed with model and process knowledge to capture physical relationships in chemical processes, achieving better robustness and reducing redundant alarms for controllable faults.

Fig. 11 shows process monitoring results on back-to-controls faults of the TE process. For fault 4, the reactor's temperature control loop responds rapidly, preventing the monitored absolute errors from exceeding their corresponding alarm thresholds. In the cases of faults 5 and 7, certain process variables may slightly deviate from the original operational condition after the fault occurrence, but the mass and gas-liquid phase transition balance relationships should remain the same after the control system takes effect. The fault alarm should be eliminated after recovery of process dynamics. Fault 5 is caused by a step temperature change of the condenser cooling water, resulting in immediate fluctuations in the system; fault 7 stems from the input pressure loss. Both faults can be observed by the unstable pressure in the reactor, stripper, and separator, disrupting the gas-liquid phase transition balance and leading to the loss of effectiveness of the established physical

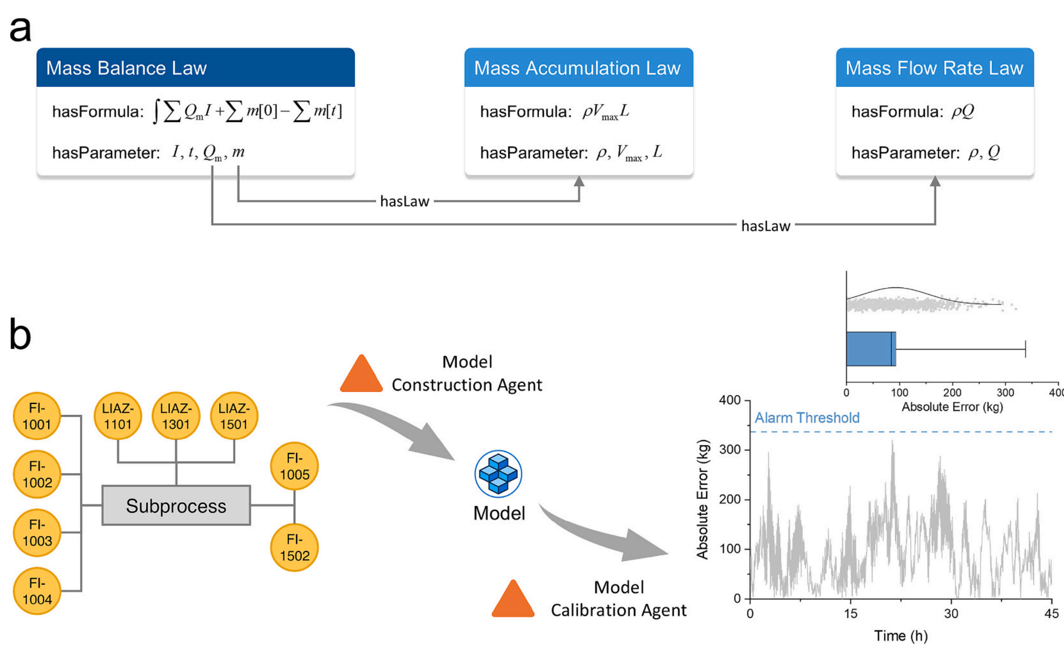

Fig. 8. Automated mass balance modelling for the TE process. (a) The mass balance law, along with subsidiary mass accumulation and flow rate laws, retrieved from the model ontology. (b) The identified subprocess and the derived mass balance model. Absolute errors under normal operational condition are computed after model calibration. Mass flow rate is indicated as Q_{m}; flow direction I; accumulated mass m; time t; density ρ; maximal volume $V_{\max}$; relative liquid level L; and flow rate Q.

model (Fig. S21). Resultant absolute errors coincide with the root causes: the mass balance absolute error remains mostly below the alarm threshold, while the phase transition balance absolute errors exceed the alarm threshold immediately after the fault occurrence and then gradually descend below the alarm threshold.

Process monitoring results for uncontrollable faults of the TE process are shown in Fig. 12. Fault 1 is caused by the step disturbance in the feed ratio of reagent A to C, which alters mass conversion dynamics across the entire system. The established mass balance model is unable to accurately predict the mass balance under the influence of fault 1, leading to significant absolute errors. Large absolute errors can also be observed in the gas-liquid phase transition balance immediately after the fault occurrence. Faults 12 and 13 mainly affect the thermodynamic behaviour of the TE process, with minimal impact on mass balance due to the relatively stable densities of streams. The TE process is operated under an unstable state because of the combined effects of the fault and the control system. The monitored absolute errors for the phase transition occasionally exceed the alarm threshold after the fault occurrence. Such differences between faults can also be interpreted by the time-series data of the flow rate of inlet stream 4. Fault 1 is the only one in which the mean flow rate varies and leads to the constantly large mass balance error. With regard to faults 12 and 13, the mean flow rate almost maintains the same after fault occurrence despite having a slightly larger variance. Therefore, the mass balance error remains below the alarm threshold for most of the time.

These cases demonstrate automation of model assembly and model calibration against process data and construction of process monitoring workflows (see process monitoring results on other faults in Fig. S21).

3.2. A practical multi-mode dosing module case

This use case originates from a pilot plant developed in the Accelerated Manufacturing Platform for Engineered Nanomaterials (AMPLE) project by Accelerated Materials Pte Ltd. We developed ontology and knowledge graph instances for the multi-mode dosing module to demonstrate the overall implementation of the knowledge-graph based digital twin. The dosing module is the initial stage in the manufacturing process and is designed to distribute liquids from four storage tanks into the subsequent reactor module through four stream groups; it can be operated in three different operational modes: rinse mode, prime mode, and reaction mode.

3.2.1. Mass balance model development

Fig. 13(a) shows the process diagram of the multi-mode dosing module. Reagents are stored in tanks 1 and 2, while tank 3 contains a rinsing liquid used to clear system blockages. Tank 4 holds deionised water, which is used to wash pipelines or dilute reagents as needed. Each storage tank is attached with a liquid level sensor, and each stream group is equipped with three control valves, corresponding to rinse, prime, and reaction modes. Flow meters are installed in each stream group to measure the volumetric flow rate. This modular pipeline configuration enables flexible transitions between different operational modes. Fig. 13(b) displays the corresponding digital representation of the dosing module (visualised with Neo4j). The process ontology includes tanks (red nodes), streams (blue nodes), pipe manifolds (green nodes), and process variables (yellow nodes). For clarity, valve states are omitted in the visualisation, and only *hasOutletStream* relationships are plotted between unit operations and streams.

Fig. 14 shows the automated construction and application of a mass balance monitoring model for the multi-mode dosing module. Starting with the mass balance *Law*, subsidiary *Laws* are enumerated to assess their applicability to the dosing module according to their connected *Parameters* and the practically measured *Process Variables* (Fig. 14(a)). In this case, volume accumulation V and volumetric flow rate Q are measured while density ρ needs to be fitted. m and Q_m can be calculated as ρV and ρQ, respectively. A one-minute time window is adopted for monitoring the mass accumulation change. The constructed model is applied to an operational scenario where only Valve_A_Re and Valve_B_Re (as referenced in Fig. 13) are open to determine the alarm threshold. The threshold is set to the mean value plus three standard deviations. A time window of 1 min is adopted for mass balance modelling.

Subprocess identification results of different operational modes can be found in Fig. S22. In the rinse mode (Fig. 14(b)), a single subprocess is identified where tank 3 is active for rinsing all four stream groups. The mass balance model is constructed and applied to the identified

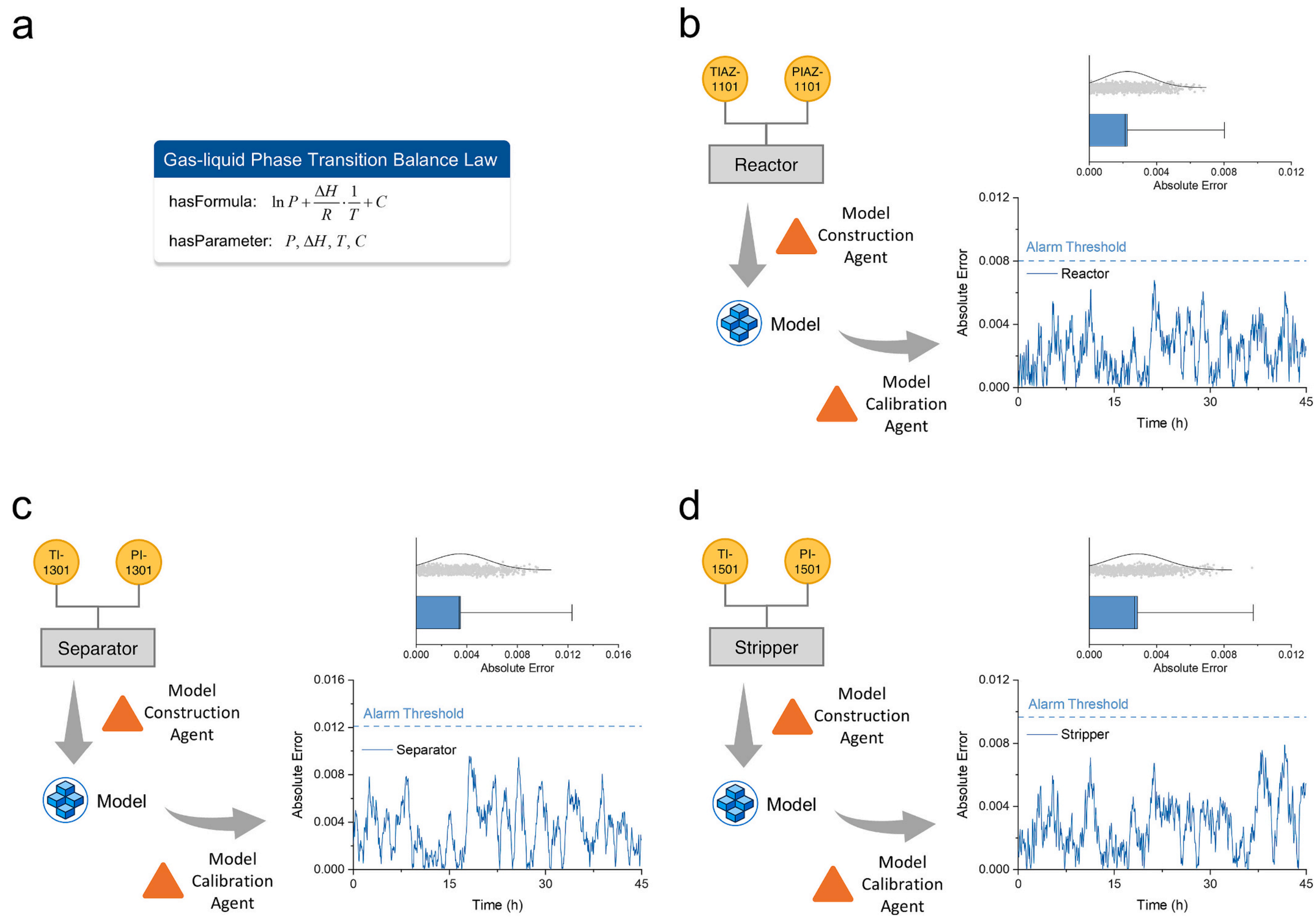

Fig. 9. Automated gas-liquid phase transition balance modelling for the TE process. (a) Retrieved gas-liquid phase transition balance law. Models are automatedly established on (b) the reactor, (c) the separator, and (d) the stripper. Respective absolute errors are calculated under the normal operational condition for identified subprocesses after model calibration. Pressure is indicated as P; enthalpy change ΔH; universal gas constant R, Temperature T; and the intercept of the Clausius-Clapeyron equation C.

subprocess by the developed software agents. The calculated absolute errors are consistently below the small alarm threshold, indicating a strong alignment between the mass balance model and the actual process dynamics. Likewise, the mass balance monitoring is applied to the prime mode, as shown in Fig. 14(c). The reaction mode (Fig. 14(d)) presents as a more complex operational scenario, including multiple subprocesses with respective level indicators associated with connected flow indicators. Tanks 1, 2, and 4 are active simultaneously for production. The absolute error results demonstrate a strong correlation between tank levels and stream flow rates, confirming the effectiveness of the ontological method to dynamically construct physical models under different operational modes.

3.2.2. *Process monitoring results*

Fig. 15 illustrates the overall anomaly detection workflow for the dosing system of the pilot plant. It can be used for both real-time and simulation tests for anomaly detection. The system can ingest data from historical data streams, either from the *Test Server* or simulated data from the *Simulator*. The *Simulator* can generate both normal and abnormal operational data based on a physical model. To generate data with noise similar to the working condition, we recorded noise' relative strength, frequencies, and patterns from practical data and reused them to generate noise for simulation data. These noise cases are scaled and then added to data from physical models to mimic real sensor data, as illustrated in Fig. S23. In this way, the generated data resembles those from practical setups, ensuring the robustness of trained AI models for anomaly detection in real plants. The use case showcases anomaly detection through two complementary approaches: one grounded in a physical model and the other driven by data analytics.

The data first goes through the pre-processing module to ensure that there are no missing entries in each data sample and each sensor reading is within a pre-defined range. From the pre-processing module, any abnormality in the data will directly trigger the post-processing module to raise an immediate alert, bypassing the anomaly detection modules. Otherwise, the pre-processed data is sent to the subsequent physics-based and data-driven anomaly detection modules. These two modules work in parallel to find anomalies from the streaming data and are combined to ensure robust and reliable anomaly detection, improving operational efficiency and reducing unexpected downtimes (Fig. S10).

Detection results are then processed in the post-processing module, where certain conditions are ensured before triggering a final alert. To minimise false alarms, an alert is only triggered when. This mechanism is demonstrated via an example in Fig. S11. Finally, the post-processed results are stored in the *MongoDB* database and visualised in the UI, which helps technical operators monitor and analyse anomalies in real time.

The deployed physics-based model is obtained as described in Section 3.2.1. This model is more reliable and adaptable when minor changes happen in the system, such as having additional sensors or changing tank-pipe connections. Besides, its computational complexity is much lower than that of the data-driven model. On the other hand, the data-driven model can cover a wider range of abnormal patterns,

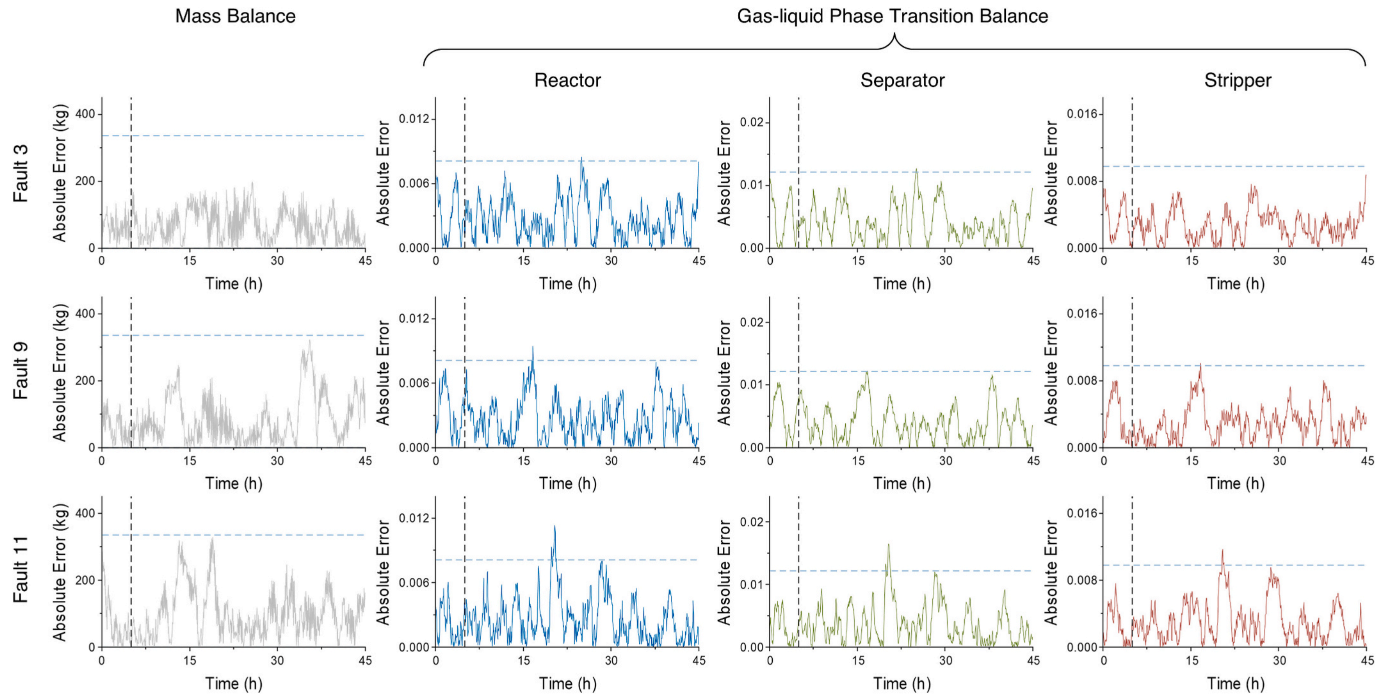

Fig. 10. Process monitoring results of example controllable faults of the TE process. The vertical dashed line indicates the fault occurrence time. Faults numbers correspond to Table 2.

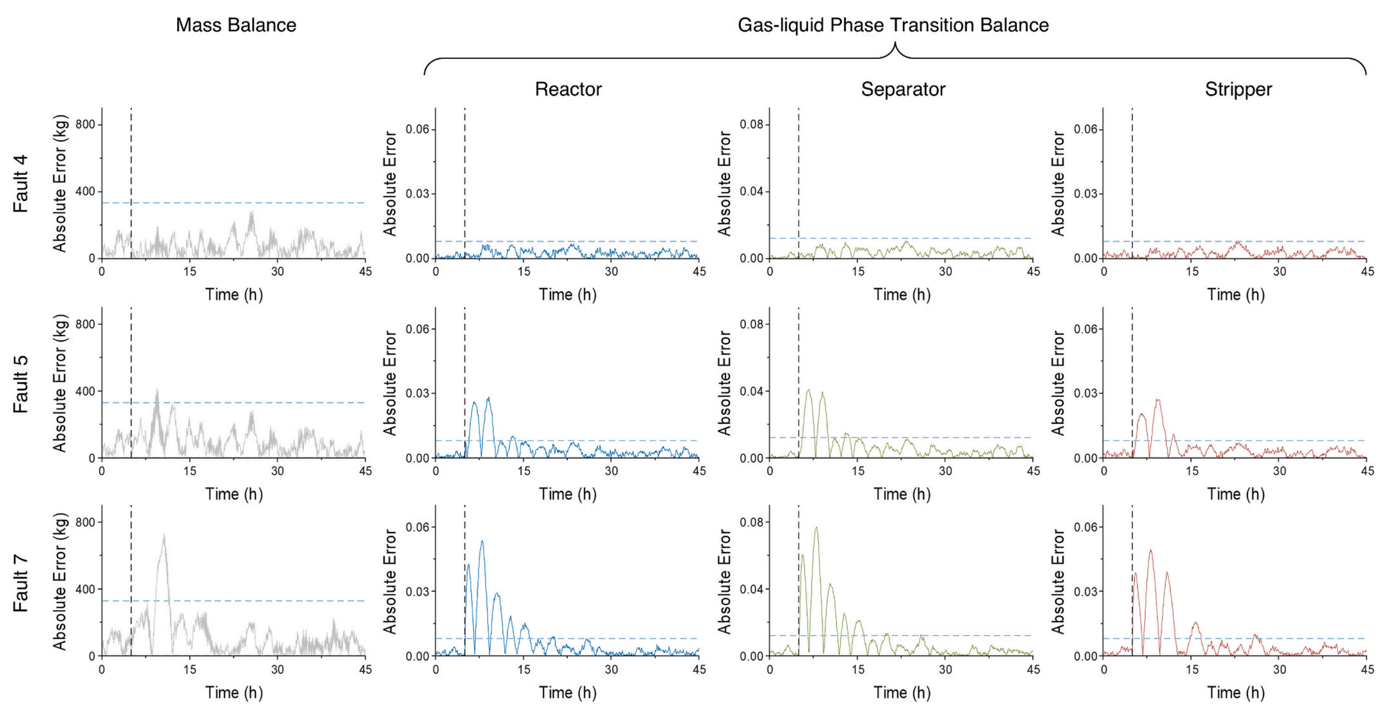

Fig. 11. Process monitoring results of back-to-control faults of TE process. The vertical dashed line indicates the fault occurrence time. Fault numbers correspond to Table 2.

including but not limited to out-of-distribution data values, mismatched configurations, and noisy sensors.

We take Autoencoder (AE) as the data-driven model to detect deviations from the norm by measuring reconstruction errors [50]. AE can learn efficient encoding and decoding processes using healthy data patterns. Fig. 16 illustrates the network architecture, as well as the procedures during training and testing. During training, AE optimises the mean squared error between the input and the reconstruction output to learn the data pattern. After that, the model would be able to reconstruct the normal data sample with minimal errors. The alarm threshold is set to a certain percentile (e.g., the 99th percentile) of all reconstruction errors from the training dataset. Based on training statistics, the anomaly detection decision threshold is set as 0.00171 to encapsulate most normal data points, as shown in Fig. S26.

From the AMPLE plant, we collected normal operational data to train an AE model. The training data consists of 28,946 samples, with each

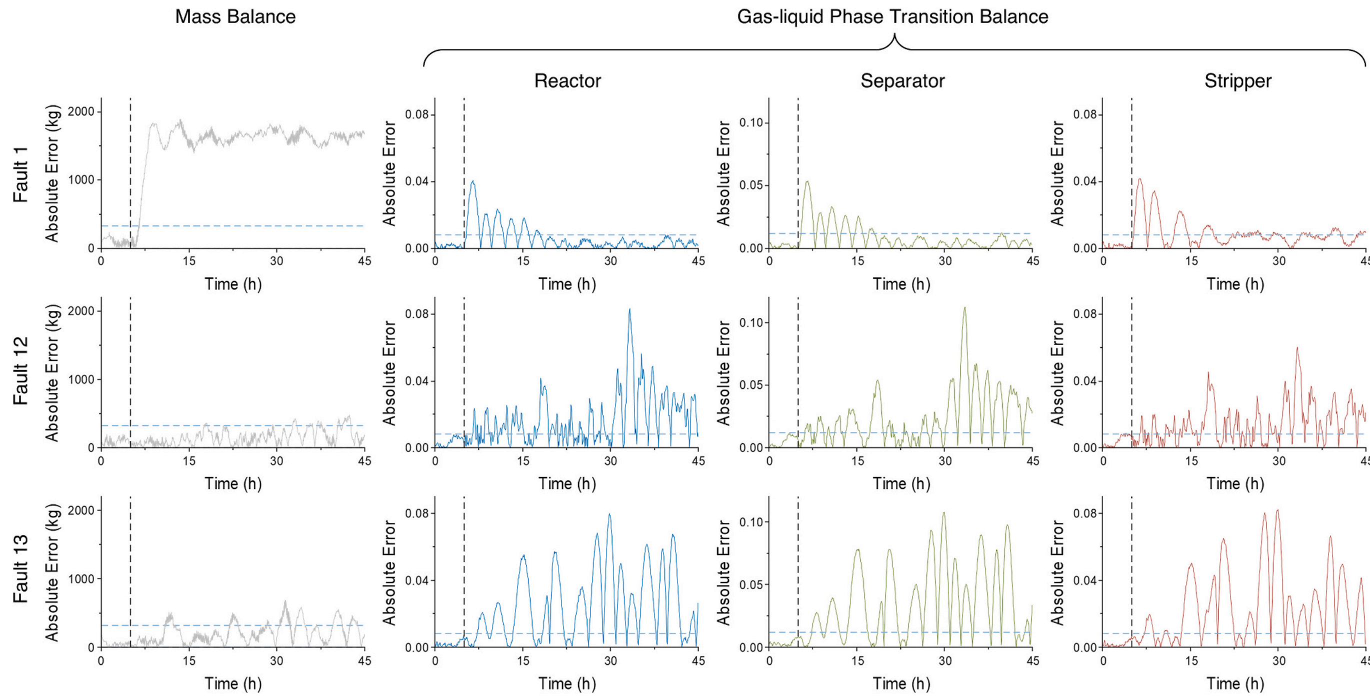

Fig. 12. Process monitoring results of example uncontrollable faults of the TE process. The vertical dashed line indicates the fault occurrence.

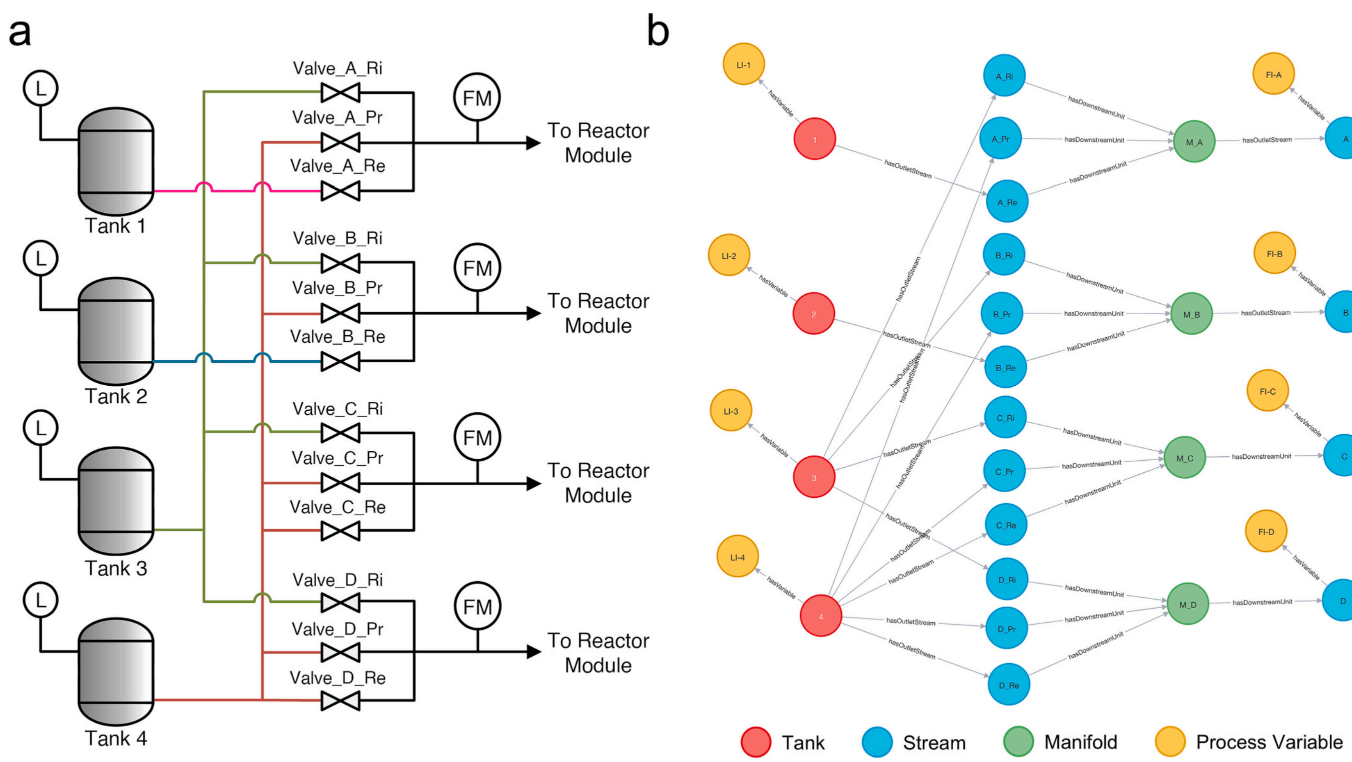

Fig. 13. (a) Process diagram and (b) digital representation of the multi-mode dosing module case.

sample having 54 parameters. The parameter set includes air flow rates (two parameters), liquid tank levels (four parameters), liquid pipe pressures (four parameters), liquid pump flow rates (four parameters), other pump-related (16 parameters), pipe temperatures (four parameters), and controlling valves (20 parameters). We provide the example time-series plot of Pipe D pressure in Fig. 17.

Due to the challenge of collecting real-world data that contains actual anomalies, we test the effectiveness of our method on simulated faults data (Fig. 18). Three types of process anomalies are created, including tank leaking, flow rate reading offset, and valve mismatch. Each test scenario is simulated for 16 min. The knowledge base of the physical model endows it with higher robustness and reliability. However, the physical model adopts a time window for process monitoring, leading to a longer response time after the fault occurrence and

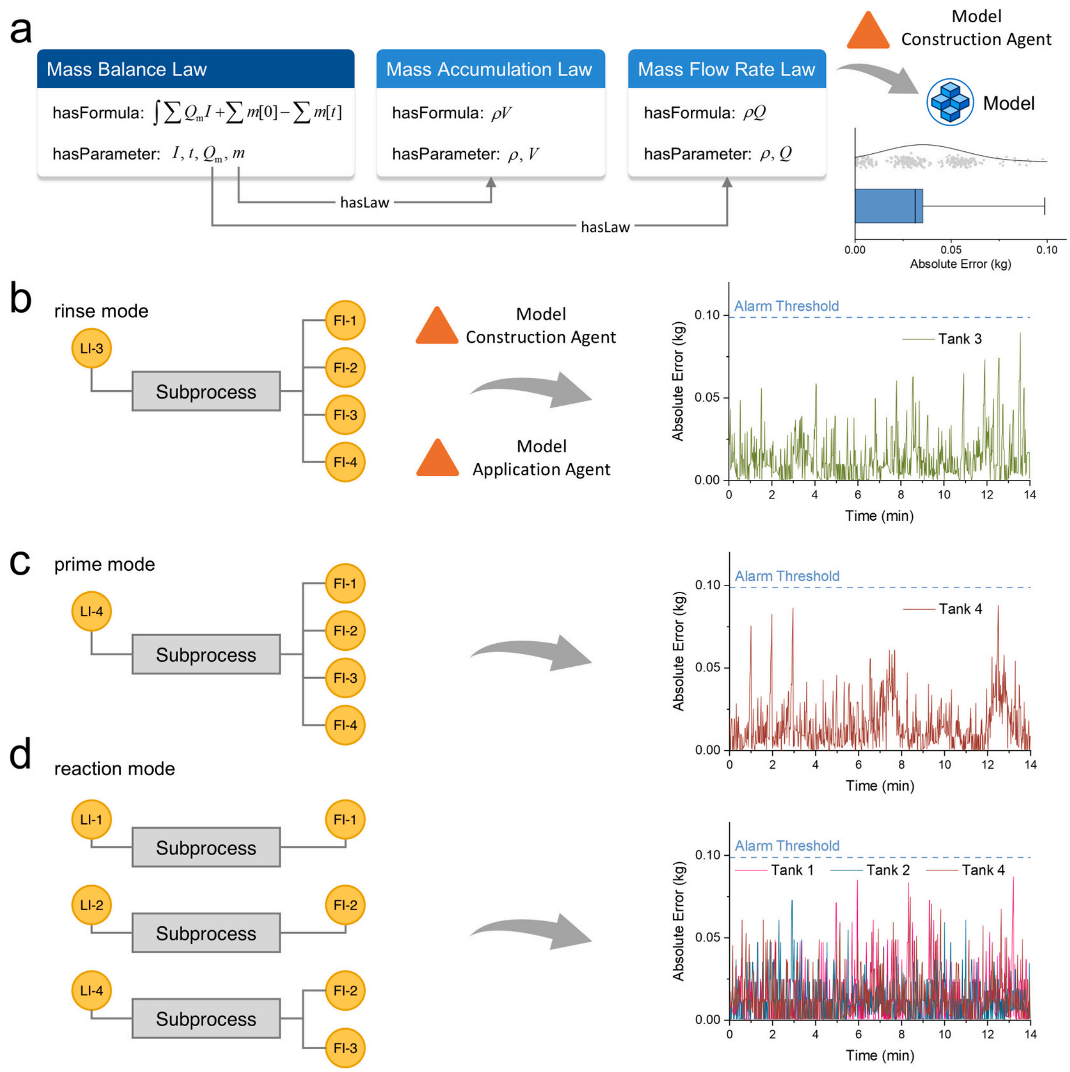

Fig. 14. Automated model construction and application using ontologies for mass balance monitoring on the multi-mode dosing module. (a) The retrieved mass balance law and associated subsidiary mass accumulation and flow rate laws by the model construction agent, with absolute errors collected in a boxplot. Respective models are derived under (b) rinse, (c) prime, and (d) reaction modes.

recovery. The AE model is more sensitive to process faults but requires collecting data for model training. Our ontological framework serves as a highly usable infrastructure to enable the application of both physical and data-driven models in chemical processes.

4. Conclusions

The digitalisation technology underpins the transition of modern chemical processes for enhanced automation, efficiency, and adaptability. It lays the foundation for intelligent manufacturing systems that are responsive to dynamic market demands and sustainability goals. To achieve this, we here proposed an ontology-based semantic framework to enable systematic modelling knowledge management, data utilisation, and model embedding as the base for applications such as real-time monitoring, predictive control, and optimised decision-making. Ontologies and agents are developed to drive process data throughout its generation, storage, processing, and application. The established platform ensures data consistency and usability, therefore enabling the deployment of both physical and AI models for chemical processes. As a demonstration, we applied the semantic framework to the anomaly detection tasks of a benchmark chemical process and a practical dosing module.

Our practical implementation demonstrates the integration of semantic technologies and a secure data streaming architecture within a digital twin framework for a process plant. Cross-domain ontologies are employed to formally represent devices, parameters, and process data, enabling a unified, machine-readable, and semantically interoperable model of the pilot plant. The data streaming architecture facilitates the transfer of measured data from the pilot plant to the user. It incorporates Keycloak for identity management, ensuring robust authentication and authorisation. Together, these components establish a reliable and interoperable data pipeline that supports advanced analytics and real-time anomaly detection.

While this study establishes a semantic framework for chemical process digitalisation using ontologies, we acknowledge specific limitations that present opportunities for further development. A primary bottleneck is the manual effort required for creating specific process ontologies for chemical plants. Future iterations could automate this by applying AI models to extract information directly from process diagrams. Furthermore, to enhance practical deployment, the framework needs to be compatible with diverse sensor vendors, cloud architectures, and data formats. This effectively necessitates close multidisciplinary collaboration across hardware, software, and chemical engineering domains. Finally, the current model ontology incorporates a finite set of physical laws; expanding this knowledge base and extending the framework to tasks beyond anomaly detection are promising.

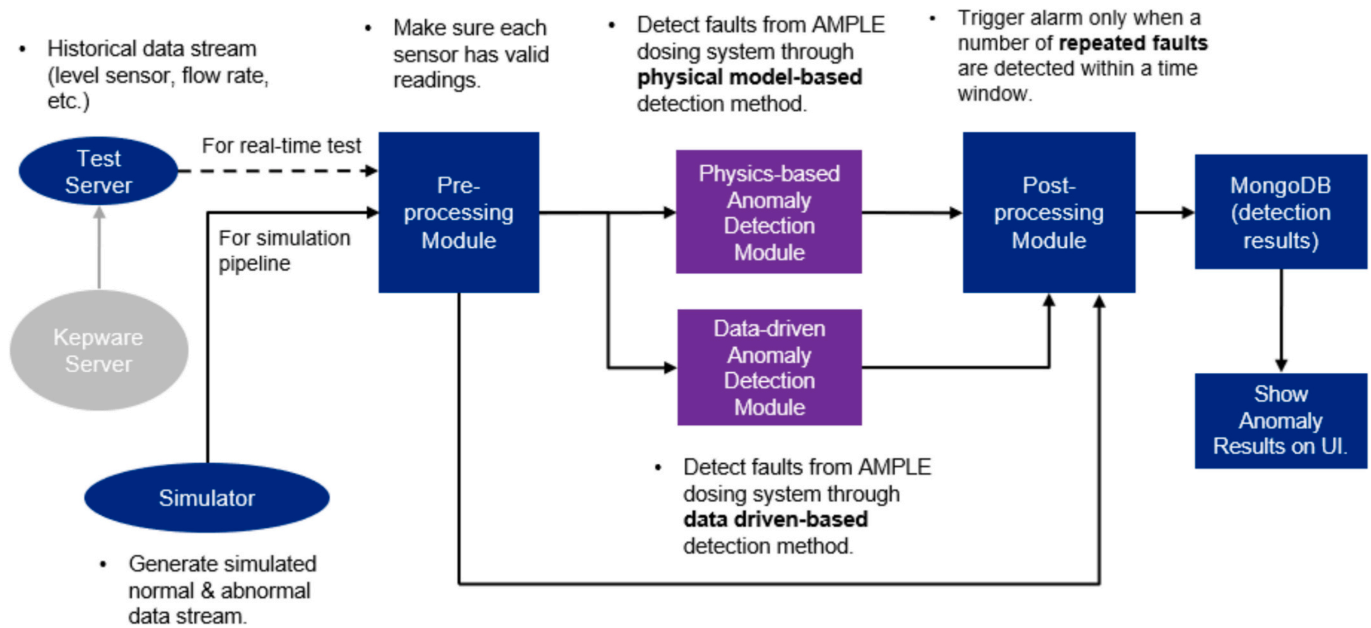

Fig. 15. An illustration of the overall process monitoring workflow, encompassing key modules of pre-processing, post-processing, physics-based anomaly detection, and data-driven anomaly detection.

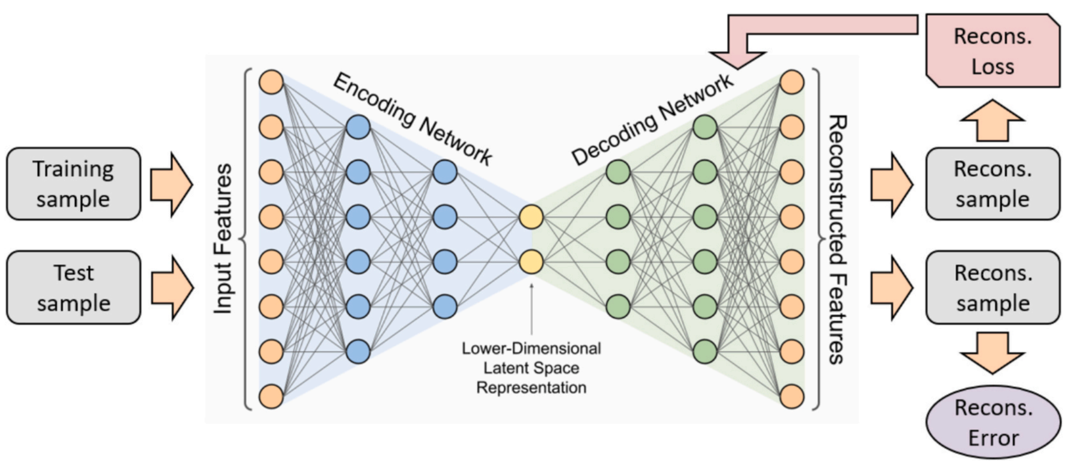

Fig. 16. An illustration of the AE model architecture for data-driven anomaly detection.

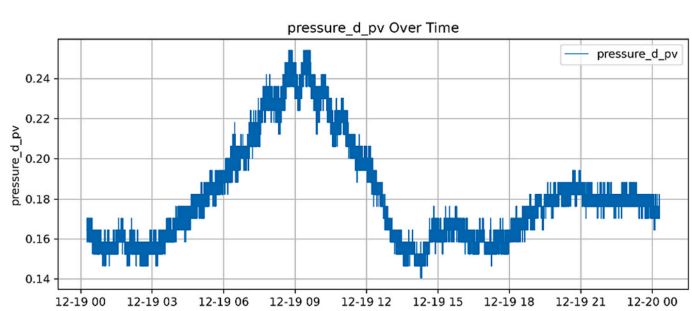

Fig. 17. Time-series plot of Pipe D pressure from the AMPLE setup.

CRediT authorship contribution statement

Shuyuan Zhang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Yong Ren Tan:** Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Cuong Manh Nguyen:** Software, Methodology, Investigation. **Dogancan Karan:** Methodology, Investigation. **Srishti Ganguly:** Software, Methodology, Investigation. **Nicholas Jose:** Methodology, Investigation, Data curation. **Mei Qi Lim:** Project administration, Investigation. **Shuqiao Guo:**

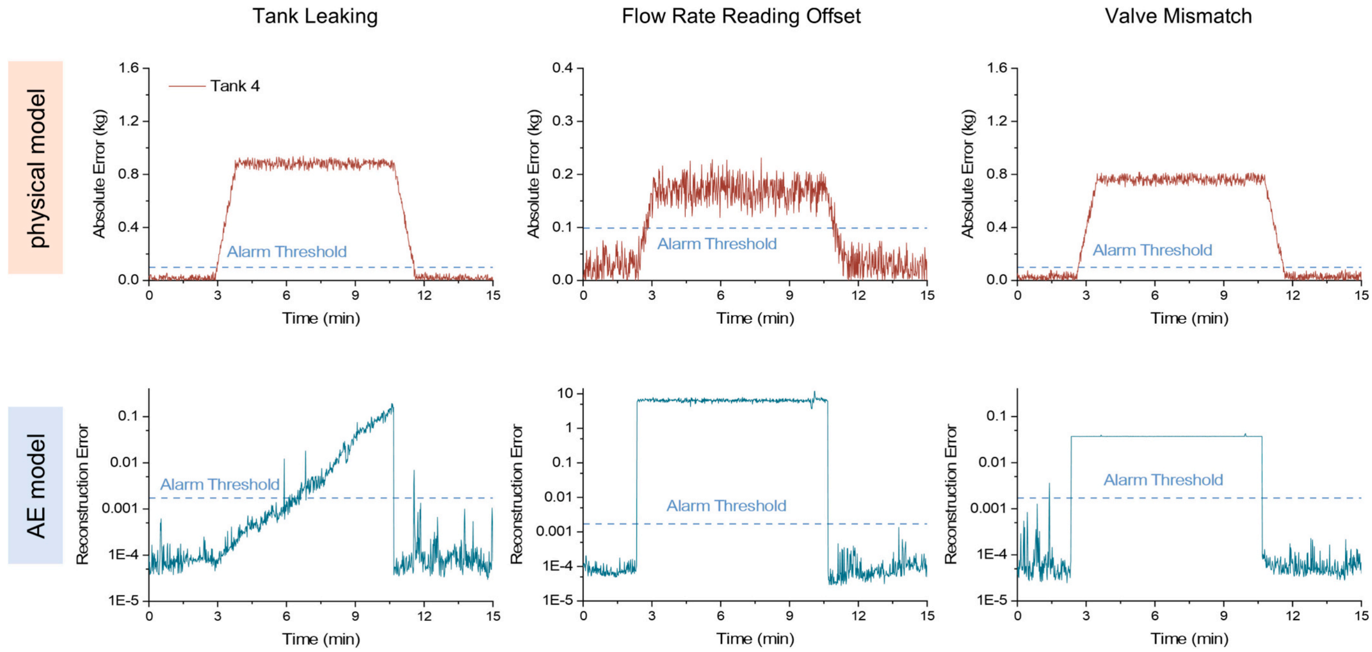

Fig. 18. Process monitoring results of the physical model and the AE model on faulty data.

Software, Methodology, Investigation. **Lianlian Jiang:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition. **Markus Kraft:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Alexei A. Lapkin:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

AAL is founder of Chemical Data Intelligence (CDI) Pte Ltd., developing commercial implementation of the model and process ontologies and knowledge graph implementations. Accelerated Materials Pte Ltd. owns the plant that was used for demonstration of the digital twin technology.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2026.174361.

Data availability

Data will be made available upon reasonable request. Core ontologies are provided under the CC-BY license and can be accessed through https://github.com/zhangsy-ryan/chemproc-onto.git. Further codes, including communication protocols, specific equipment ontologies and models containing proprietary information, could be shared upon specific and reasonable requests.

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